

Overage, Pricing & PO Stories (Build Spec)

A. Overage mechanics (line-item first)

A1) Define a line-item Base Price (“Obase” in Oasis) and a Sell Price

Relevant Client Comments

Client Context: Amanda: OBase is the manufacturer base cost (auto-loaded from price lists/APIs where available). Anything above OBase in the sell price is treated as overage; manufacturers typically pay a fixed share of overage (e.g., 90%).

Clarification: Default OBase from manufacturer price lists or API; allow manual override per line. Overage share is set at the manufacturer/factory level (never per SKU).

Story

As a sales rep, I want to set a base price per line so anything above this price is overage, so that overage earnings are automatically computed.

Acceptance Criteria

- Quote & Order Line item supports fields: `base_price` (refers to the qty price currently in our system), `sell_price` (new optional field if the rep wants to sell at a different price.)
- System computes:
 - `overage_amount = max(sell_price - base_price, 0)`
 - `overage_commission = overage_amount * rep_overage_share` (e.g., 0.90)
 - `base_commission = base_price * commission_rate_base` (this is how we currently calculate commissions)
 - `total_commission = base_commission + overage_commission` (This would be the new standard commission field, if overage is 0 then it doesn't change)
- Overages compute **only** when item is priced at the **highest eligible commission band** (see A5).
- `rep_overage_share` is set on the factory level and is not allowed to be over 100%

UI/UX

- Add a “sell price” and “Overage Amt” next to “Unit Price” on the line item, if the user clicks an “Overage” toggle on the quote. That way the UI doesn't always get crowded with these extra fields if they are not needed for the quote.
- Orders have the same toggle, and it's on by default if created from a quote with overage
- The standard commission field should update with the overage added to get the total commission

Edge

- If `sell_price <= base_price`, then `overage_amount = 0`.

Screen shots from Oasis (just for reference, not to duplicate)

	Price	Comm	Earn	Rate
Sell:	\$100.00	\$5.00	\$5.00	5%

	Price	Comm	Over	Earn	Rate
OBase:	\$100.00	\$5.00		\$5.00	5%
Sell:	\$150.00	\$5.00	\$45.00	\$50.00	33.33%

A2 – Description visibility like in Oasis so reps know what they’re working with

As a sales rep, I want to toggle between showing product numbers and product descriptions (max 255 characters) **on each individual quote, order, or invoice**, so I can adjust visibility depending on the context of that document. As an admin, I want to set a default display mode for all documents, so users don’t have to change it unless they want to.

Relevant Client Comments

Client Context: From usage: Reps switch views to keep quotes readable for customers and internal teams. In dense, multi-manufacturer quotes, description view improves clarity when sharing.

Clarification: Persist a per-document toggle for product number vs description; admin sets default. Invoices inherit the order’s setting.

User Stories

1. Per-document toggle

- As a user, I want a toggle on each **quote, order, and invoice screen** that lets me switch between product number and description, so I can view the most helpful detail for that document.
- Acceptance Criteria:

- Toggle is present on Quotes and orders (Invoices inherit the setting from the Order they are attached to)
- Options: Product number or Description.
- When “Description” is selected, truncate at 255 chars with “...”.
- Toggle state applies only to the document I’m currently viewing (not global).

2. Persist toggle per document

- As a user, I want the toggle choice saved with the document, so that when I reopen that quote/order/invoice, it shows the last selected mode.
- Acceptance Criteria:
 - Open a quote or order, change the toggle, navigate away
 - On reopen, see that toggle persisted

3. Admin default setting

- As an admin, I want to configure whether new documents default to showing product number or description, so users see the organization’s preferred view.
- Acceptance Criteria:
 - Org-level setting: `default_line_item_display_mode`.
 - When a new quote/order/invoice is created, initialize `line_item_display_mode` to this setting.
 - Users can still override per document.

Example from Oasis:

MEG0125-63910 (Base Counts (Options))

Bid: Initialed Base: \$84,090.14 \$8,434.22 \$8,434.22 10.03%
 Quote Status: Prebid/Design OBase: \$84,090.14 \$8,434.22 \$8,434.22 10.03%
 Sell: \$241,860.89 \$39,831.72 \$4,852.16 \$44,683.88 18.8%
 Level1: \$262,661.21 \$8,434.22 \$11,854.26 \$20,288.46 7.73%
 Level2: \$277,646.82 \$8,434.22 \$16,885.34 \$25,319.56 9.12%

For... RADate: /MM/YYYY-HH:MM:AA
 BidDate: 07/26/2025 101:44 PM
 Exp. Date: 09/24/2025

Items	Customers	Manufacturers	Notes	Part	Descripti...	Note	OBASE	Base	Bas...	%	Sell Pr...	Sell C...	Level 1	MFG Tar...	%Ove...
1	Inf...				7% OVERAGE ON SLG PORTURES										
2	Inf...				LEVEL 1 HAS 5% ADDED FOR CR & TCHO, CR CAN TAKE 5% OFF										
3	Inf...				LEVEL 2 HAS 12% ADDED FOR CR CORRECT (CHICAGO DISTRIBUTOR - QUOTING DOMINON)										
4	Inf...														
5	NO...				HEALTH DEPARTMENT BUILDING:										
6	99	2X2	SLG		TBSC 32 LRAE G1 POK-AV (25K, 32W/3750LM)		\$44.00	\$44.00	10%	7	\$47.31	(15.6%)	\$51.42		80
7	99	2X2	SLG		ADDER FOR AIO-ON CONTROL INSTALLATION FOR ABOVE PRODUCT		1...	10%			\$10.00	0%	\$10.87		90
8	1,007	2X4	SLG		TBSC 24 LRAE G1 POK-AV (35K, 45W/3400LM)		\$60.00	\$60.00	10%	7	\$64.52	(15.6%)	\$70.13		80
9	1,007	2X4	SLG		ADDER FOR AIO-ON CONTROL INSTALLATION FOR ABOVE PRODUCT		1...	10%			\$15.00	0%	\$16.30		90
10															
11	22	4" STRIP	SLG		TSS 4 30/40/50 G1 POK-AV (25K, 32W/4180LM)		\$39.00	\$39.00	10%	7	\$41.94	(15.6%)	\$45.59		80
12	15	4" STRIP	SLG		TSS-30CVR		\$1.00	\$1.00	10%	7	\$1.06	(15.6%)	\$1.12		90
13	7	4" STRIP	SLG		TSS-CHAM		\$8.00	\$8.00	10%	7	\$8.60	(15.5%)	\$9.35		90
14															
15	4" STRIP	SLG			SAC-1810		\$9.00	\$9.00	10%	7	\$9.60	(15.6%)	\$10.32		90
16	7	4" STRIP	SLG		TSS-INDOR		\$1.00	\$1.00	10%	7	\$1.06	(15.6%)	\$1.12		90
17	2	4" STRIP	SLG		RC-TSS G1		\$2.00	\$2.00	10%	7	\$2.22	(15.7%)	\$2.51		90
18	15	4" STRIP	SLG		ADDER FOR AIO-ON CONTROL INSTALLATION FOR ABOVE PRODUCT		1...	10%			\$10.00	0%	\$10.87		90
19															
20	27	8" STRIP	SLG		TSS 8 60/80/100 G1 POK (25K, 64W/8320LM)		\$74.00	\$74.00	10%	7	\$79.57	(15.6%)	\$86.40		80
21	27	8" STRIP	SLG		TSS-CHAM		\$8.00	\$8.00	10%	7	\$8.60	(15.5%)	\$9.35		90
22	8" STRIP	SLG			SAC-1810		\$9.00	\$9.00	10%	7	\$9.60	(15.6%)	\$10.32		90
23	8" STRIP	SLG			TSS-INDOR		\$1.00	\$1.00	10%	7	\$1.06	(15.6%)	\$1.12		90
24	2	8" STRIP	SLG		RC-TSS G1		\$1.00	\$1.00	10%	7	\$1.22	(15.7%)	\$1.57		90
25															
26	33	4" WRAP	SLG		TR 4 35/45/55 G2 POK-AV (35K, 24W/3800LM)		\$74.00	\$74.00	10%	7	\$79.57	(15.6%)	\$86.40		80
27	33	4" WRAP	SLG		ADDER FOR AIO-ON CONTROL INSTALLATION FOR ABOVE PRODUCT		1...	10%			\$10.00	0%	\$10.87		90
28															
29	14	6" DOWN	SLG		DRC 6R L520 G1 POK TAM (33K, 14W/2100LM)		\$27.00	\$27.00	10%	7	\$29.02	(15.5%)	\$31.55		90

Save and close Close Print

A3) Set % Overage on Selected Lines

Relevant Client Comments

Client Context: Amanda: Heavy daily use of typing a percent on a line (e.g., '7') to auto-recalculate sell price; also uses a bulk apply to set the same percent across rows.

Clarification: Support BOTH inline per-line % overage entry and a bulk 'Set % Overage' action with preview. Use formula: $sell = base_price / (1 - p)$.

Story

As a sales rep, I want to select **one, multiple, or all** line items in a quote/order and set a **% overage** for the selection in one action, so I can price quickly while still having fine-grained control.

Acceptance Criteria

- **Selection scope**
 - Reuse the existing line-item selection workflow (single, multi-select, “Select all”).
- **Action**
 - New bulk action: **“Set % Overage”** (available when ≥ 1 line selected).
- **Calculation (applies per selected line)**
 - Input: p as percent (e.g., 7 for 7%). Validate $0 \leq p < 100$.
 - Compute: $sell_price = round(base_price / (1 - p/100))$.
 - Recompute per line: `overage_amount`, `overage_commission`, `total_commission`,
- **Eligibility**
 - **Skip** lines with overage disallowed `base_price` is missing or zero, or $sell_price < base_price$
- **Preview & confirmation**
 - Before applying, show a modal summary:
 - **net total change** in document `sell_price`, `overage_commission`, and `total_commission`.
- **Result & feedback**
 - Apply the formula to all eligible selected lines in one transaction.

UI/UX

- Use the existing line-item selection (checkboxes + “Select all”).
- Add **“Set % Overage”** to the bulk actions toolbar (same pattern as other multi-line functions).
- Clicking it opens a compact modal with:
 - Numeric input for % overage.
 - Inline validation (e.g., “Enter 0–99.99”).
 - Preview block (`sell_price`, `overage_commission`, and `total_commission`).
 - “Apply” / “Cancel”.

- After apply, show success toast + “Undo” link.
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A6) Commission bands / levels (enforce eligibility)

Relevant Client Comments

Client Context: Amanda: Commission bands are price windows below OBase; each window yields a different commission rate. Only when $\text{sell} \geq \text{OBase}$ is overage allowed (top band).

Clarification: Compute $\text{delta_pct} = (\text{OBase} - \text{sell}) / \text{OBase}$ and choose the band whose $[\text{min_delta_pct}, \text{max_delta_pct})$ contains delta_pct . If $\text{sell} \geq \text{OBase}$, use the top band and set $\text{overage_allowed} = \text{true}$; otherwise false.

Story

As a sales rep, I want commission “bands” (e.g., 10% / 8% / 6% / 4%) tied to distinct base price windows below the sell price.

Acceptance Criteria

- Each SKU has one or more `commission_band` entries: `{band_rate, band_base_price, overage_allowed}`. This should be implemented on the product screen.
- If user selects a lower band (e.g., 4–8%), system sets `overage_allowed = false`.
- UI “Price Lookup” on the order or quote shows all bands and indicates which allow overage.

Data Model

- `CommissionBand: sku_id, band_rate, band_base_price, overage_allowed:boolean`.
-

A7) Zero-commission items

Relevant Client Comments

Client Context: Amanda: Some SKUs are truly non-commissionable (0%). They shouldn’t count toward target effective commission and shouldn’t trigger or suggested overage.

Clarification: Mark lines `commissionable=false` with `commission_rate_base=0`; default `overage_allowed=false` on such lines. Allow an org/manufacturer policy override only if explicitly enabled.

Story

As a sales rep, I want items with 0% commission to be supported and excluded from overage rollups that could skew results.

Acceptance Criteria

- Line item can store `commission_rate_base = 0`.
- Overages can still apply **only** if band allows and manufacturer permits (else 0).
- Any “target effective commission” helpers should ignore these lines in auto-calc suggestions.

A8) Line-item vs total-BOM overage mode (edge case, less important)

Relevant Client Comments

Client Context: Amanda: A manufacturer allowed overage only when the total BOM sell exceeded the sum of OBase for that manufacturer—rare edge case; default is per-line evaluation.

Clarification: Add `overage_mode='line'|'bom_threshold'`. In BOM-threshold mode, permit overage only if `sum_sell > sum_obase`; allocate the overage pool across positive-delta lines proportionally and disable per-line overage inputs.

Story

As an admin, I want a manufacturer policy switch: (a) overage checks per line (default) or (b) only when total BOM exceeds all Base Price totals (rare), so we can handle outliers.

Acceptance Criteria

- Manufacturer config: `overage_mode = 'line' | 'bom_threshold'`.
- For `bom_threshold`, compute:
 - `sum_obase = Σ base_price where overage_eligible`
 - `sum_sell = Σ sell_price`
 - Allow overage only if `sum_sell > sum_base_price`. (Line deltas then split proportionally.)
- Default all manufacturers to `line`.

B. Multi-level pricing & distribution

B1) Multi pricing levels (Sell, L1, L2, L3)

Relevant Client Comments

Client Context: Amanda: Quotes distributors and contractors off the same quote. Creates L1 by copying Sell with +10%, sometimes a special L2 adjustment, then batch-sends to recipients with different assigned levels.

Clarification: Price levels (Sell/L1/L2/L3) are customer-facing and separate from commission bands. Recipients see only their assigned level; internal users can view all levels and totals. Include bulk transforms, per-line locks, and an audit trail.

Story

As a sales rep, I want multiple price columns (Sell = distributor cost; L1/L2/L3 = adjusted), so I can quote distributors and contractors appropriately.

Acceptance Criteria

- Quote has `sell_price`, `level1_price`, `level2_price`, `level3_price`.
- Bulk transform: “Copy Sell → Level N with +X% markup.”

UI/UX

- “Price Levels” panel showing per-level totals; show who gets what.
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B2) Assign price level per customer (per quote)

Relevant Client Comments

Client Context: Amanda: Needs to assign a price level per recipient so each gets the correct column (Sell/L1/L2/L3) when batch sending.

Clarification: Store price_level per recipient; send/preview uses that column; block send until every recipient has a valid level and non-blank prices.

Story

As a sales rep, I want to assign each customer on a quote to a specific price level (Sell/L1/L2/L3), so sending one batch uses the correct price for each.

Acceptance Criteria

- Customer rows on the quote have `price_level` = {Sell|L1|L2|L3} with default = Sell.
- When previewing/sending, each customer’s PDF/email uses their assigned column.

Edge

- If a chosen level is blank, block send for that customer with an inline fix prompt. (Matches the “no price shows when L1 is blank” behavior.)
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B3) Batch send with checkbox selection

Relevant Client Comments

Client Context: Amanda: Selects many recipients and sends tailored quotes at once; each recipient sees the price column assigned to them.

Clarification: Provide batch send with per-recipient level assignment and validation; internal preview supports 'All Levels' for QA.

Story

As a sales rep, I want to select many customers and send tailored quotes in one action, so that each receives the right price level without manual exports.

Acceptance Criteria

- Multi-select list of customers with their assigned price level shown.
 - Emails are generated with the respective price column and sent in one batch.
-

B4) Internal notes on quotes

Relevant Client Comments

Client Context: Amanda: Writes internal notes like '7% overage on SLG fixtures; L1 +8% for Graybar & Echo' to document decisions.

Clarification: Internal notes must be timestamped, author-tagged, persisted, and never customer-visible; include in history.

Story

As a sales rep, I want to add free-text internal notes (e.g., “7% overage on SLG fixtures; L1 +8% for Graybar & Echo”), so future adjustments are traceable.

Acceptance Criteria

- Notes support timestamps and author; shown in a “Notes” panel and included in history (not customer-visible).
-

C. Agent Purchase Orders (Lighting)

C1) Generate Agent PO from Distributor PO

Relevant Client Comments

Client Context: Amanda: Many lighting manufacturers require an Agent PO in addition to the distributor PO; Agent PO shows commission/overage owed per line.

Clarification: Provide a wizard to generate Agent PO bundled with distributor PO; include columns for sell, distributor_net, commission_rate, commission_amount, overage_amount, total_due_to_agency.

Story

As an agent, I want the system to generate an Agent PO that mirrors the distributor PO plus a commission/overage breakdown, so manufacturers can verify what's owed.

Acceptance Criteria

- Upload or reference the distributor PO.
- System produces Agent PO with columns: qty, part_no, sell_price, distributor_net, commission_rate, commission_amount, overage_amount, total_due_to_agency (commission + overage).
- Both documents (distributor PO + Agent PO) are packaged to send to manufacturer.

UI/UX

- “Create Agent PO” wizard from an order; final preview shows line and totals blocks.
-

C2) PO match & variance check

Relevant Client Comments

Client Context: Amanda: Manufacturers sometimes change commission after submission; system must detect and track variances and reasons.

Clarification: Automate comparison (distributor PO, Agent PO, and statements); flag mismatches; require 'variance reason'; report matched vs mismatched lines.

Story

As an agent, I want an automated comparison between distributor PO, Agent PO, and (later) commission statements to detect variances (e.g., 10% vs 2.5%), so discrepancies are caught early.

Acceptance Criteria

- On creation, validate that line totals and pricing align; flag any differences.
 - Store a “variance reason” when manufacturer replies with changed commission.
 - Report shows matched vs. mismatched lines and expected vs. paid commission.
-

D. Special pricing modes & inputs

D1) Quantity/volume/weight pricing

Relevant Client Comments

Client Context: From notes: Southwire prices by foot/pound with frequent rate updates; Electrotech has volume tiers.

*Clarification: Support sku_pricing_mode: unit/by_foot/by_pound/tiered_qty with explicit examples; extended price = rate * measure; admin UI for frequent rate refresh with audit.*

Story

As a sales rep, I want SKUs that price by foot/pound or volume tiers to compute extended prices correctly and tolerate frequent price updates.

Acceptance Criteria

- sku_pricing_mode: unit, by_foot, by_pound, **or** tiered_qty.
 - For mode by_foot/by_pound, **extended price** = rate * measure.
 - Admin UI to refresh rates frequently (audit changes).
-

D2) Check-level discounts & expected-commission exceptions

Relevant Client Comments

Client Context: Some factories apply a remittance-level “check discount” (haircut) to the commission check, and some pre-specify “expected commission” per line (e.g., 2% instead of the standard 10%) for special pricing.

*Clarification: Support BOTH: (1) remittance-level check adjustments on manufacturer statements, and (2) per-line expected-commission overrides (rate or amount). Reconciliation **applies** line overrides first, **then** check-level adjustment proportionally, and only then computes variance.*

Story

As an **ops user**, I want to capture factory **expected-commission overrides** by line (rate or amount) **and** any **remittance-level check adjustments**, so reconciliation matches the factory’s intent even when the check is discounted.

Acceptance Criteria

- **Line overrides:** expected_commission_rate_override (decimal), expected_commission_amount_override (currency), override_reason (text).
- **Remittance adjustments (Manufacturer Statement):** support check_adjustments[] with fields:
 - type ∈ {percent, amount}, value, reason_code, applied_scope ∈ {entire_statement | manufacturer | po | order}, note, created_by, created_at.
- **Reconciliation precedence:**
 - 1) Compute expected commission from base rate.
 - 2) If a line override exists, use it (if both rate and amount are provided, **amount takes precedence**).
 - 3) Apply any check_adjustments percent/amount **proportionally** across the affected lines within the scope.

4) Compare the adjusted expected commission to actual (from statement) → **variance**.

- **Validation:**

- Cannot set both rate **and** amount overrides on the same line unless policy allows; if both present, amount wins.

- A negative or >100% rate override is rejected (unless an admin override policy is on).

- **UI/UX:**

- Badge lines with overrides (“Expected commission overridden”).

- Statement screen lists check adjustments with reason, scope, and who/when; supports add/edit with audit.

- Hover/tooltips show how each line’s expected commission was derived (base → override → check adjustment).

- **Imports:** CSV/EDI import may populate line overrides and check adjustments; conflicts are flagged for review before posting.

Data Model

- **Quote/Order Line:** `expected_commission_rate_override: decimal,`
`expected_commission_amount_override: money, override_reason: text.`

- **ManufacturerStatement:** `id, period_start, period_end, manufacturer_id,`
`check_adjustments: [{type, value, reason_code, applied_scope, note,`
`created_by, created_at}].`

- **Reconciliation (computed):** `expected_commission_pre_adjustment,`
`expected_commission_post_adjustment, variance_amount, variance_reason.`

Examples

- **Ex 1 — Line override only:** Sell \$1,000; base 10% → \$100. Factory sets override 2% → **expected \$20**. No check adjustment ⇒ variance uses \$20.

- **Ex 2 — Check adjustment only:** Two lines total expected \$120. Factory applies **-10%** check adjustment ⇒ **expected \$108**; allocate -10% to each line.

- **Ex 3 — Both:** Line override → \$20; then **-10%** check adjustment ⇒ **\$18** expected for that line; variance uses \$18.

D3) Sales order # vs purchase order # matching

Relevant Client Comments

Client Context: From notes: Some manufacturers return only a sales order # (no PO #), making matching/reconciliation harder.

Clarification: Store both `customer_po_number` and `manufacturer_sales_order_number`; matching logic should accept either and surface confidence flags.

Story

As an ops user, I want to match Manufacturer Sales Order Numbers when POs aren’t provided, so we can still reconcile orders/commissions.

Acceptance Criteria

- Order record holds: `customer_po_number`, `manufacturer_sales_order_number` (both optional).
 - Matching logic uses either identifier with confidence flags.
-

E. Targeting & guidance (quality of life)

E1) Effective commission target helper

Relevant Client Comments

Client Context: Amanda: Users target an effective commission (e.g., 18%) and adjust overage by line to reach the goal while skipping 0% lines.

Clarification: Provide a target effective commission helper that proposes % overage distribution across eligible lines; user can accept per-section or per-line.

Story

As a sales rep, I want a target effective commission (e.g., 18%) and receive suggestions on where to add overage (by line), so I can hit the goal without manual trial-and-error.

Acceptance Criteria

- Input: `target_effective_commission_rate`.
 - System proposes a distribution of `%overage_input` across overage-eligible lines (ignores zero-commission lines) and shows projected totals.
 - User can accept all, per-section, or per-line.
-

F. Rare order-value commission scaling (toggle per mfr) (edge case, lower priority)

F1) Stepped commission on total order value (rare but requested)

Relevant Client Comments

Client Context: From notes: Some manufacturers use stepped commission tied to total order value thresholds (rare but requested).

Clarification: Add `order_commission_policy='standard'|'stepped_total'`. When `stepped_total`, compute order-level effective commission using steps and allocate back to lines proportionally (feature-flag).

Story

As an admin, I want to configure stepped commission rates that kick in when the **total** order hits thresholds (e.g., $> \$X \rightarrow$ higher rate), so we can support certain manufacturers.

Acceptance Criteria

- `order_commission_policy = 'standard' | 'stepped_total'`.
- When `stepped_total`, compute order-level effective commission using the defined steps; allocate back to lines proportionally for reporting.

Note

This is distinct from overage and is less common; keep behind a feature flag.
